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## 1.0 Purpose

The purpose of this overview is to provide information on overview aspects of the processes being carried out by the VaPro Coating Line 6 Machine and the products being used and manufactured by them.

## 2.0 Scope

This is an information document and applies to the processes carried out by the VaPro Coating Line 6 Machine and the products / raw materials being used by their processes.

## 3.0 Definitions

| | | |
|---|---|---|
| *VCL* | - | *VaPro Coating Line* |
| EM | - | Equipment Module |
| *WIP* | - | *Work in Progress* |
| *70% IPA* | - | *70% Isopropyl Alcohol* |

## 4.0 References

| |
|---|
| N/A |

## 5.0 Roles and Responsibilities

Production representative for the VaPro Coating Line 6 ensures that the information in this document is current.

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## VAPRO COATING LINE 6 (VCL6)

This machine is a three (3) person operation.

The operator(s) load uncoated catheters into the manual infeeds, a pick and place unit places them on a carosel, they then get loaded onto carriers.

When 12 carriers are fully loaded they are sent through the machine, this process includes, plasma treating the catheters in a vacuum chamber, applying a top coat, purge excess top coat from the lumen (purge step is not applicable for unpunched catheters), flips the carriers 180°, and cure the top coat in 1 of the 2 pairs of cure stations.

Once the catheters are cured and dryed the carriers are placed on the outfeed belt, the catheters are stripped from the fork and loaded into grey product totes. Catheters are randomly taken from the product totes for in process sample testing and quality testing.

Once the catheters are stripped from the carrier the empty carrier returns back through the carrier test system.

VCL6 will be defined by two major machine sections:

1. Unit 1- Feeding
2. Unit 2- Process

The two units work independently of each other, but the safety (e-stop) interferes with both.

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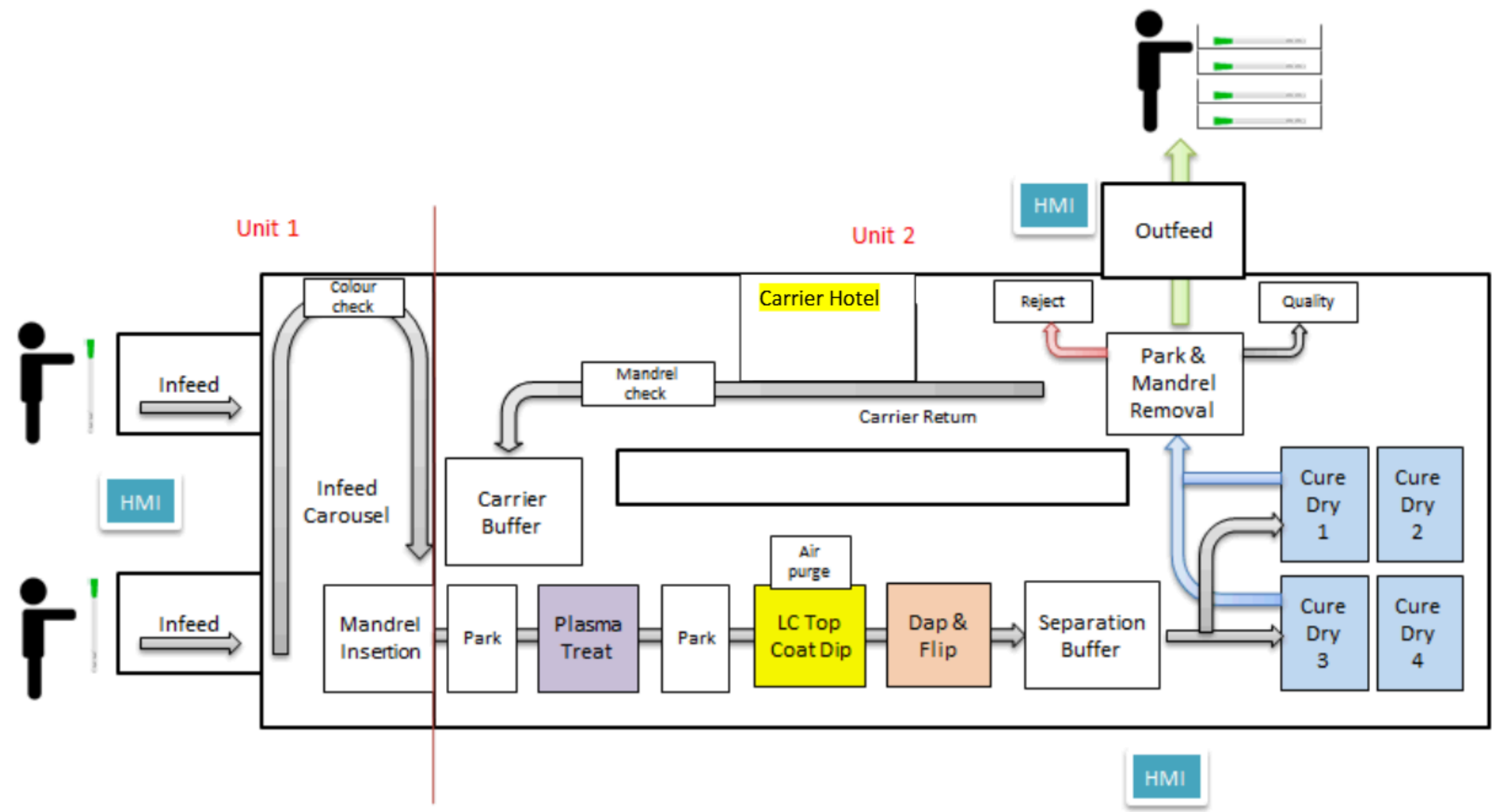


Unit 1 Unit 2

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|  | **EMERGENCY STOP**<br>In the event of an emergency push this button to STOP the machine. |

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## INTRODUCTION

| Suppliers | | Operation | | Customers |
|---|---|---|---|---|
| Bawel/ Inmo-Tech | → | VaPro Coating Line 6 | → | NGP1 |

| **Responsibilities:** | |
|---|---|
| Team Leader and Training Coordinators: | Manufacturing Instruction Implementation. |
| Operators: | Adherence to the Manufacturing Instruction and Standard Work. |

**Learning Objectives:**

At the end of this training activity the Trainee will be able to:

- Operate the VaPro Coating Line 6 in conjunction with the Work order in a safe and efficient manner.
- Operate the machine to produce a minimum target set by the Team Leader

| Training Method: | Equipment: |
|---|---|
| Training Method will involve:<br>• Job Talk<br>• Practical Demonstration<br>• Stamina Development | Listed below are the equipment requirements necessary for this operation:<br>VaPro Coating Line 6<br>Gloves<br>Isopropanol 70%<br>Lint free wipes |

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## MATERIALS FOR VAPRO COATING LINE 6 (VLC6)

Catheter Subs - Are received from Bawel / Inmo-Tech in a blue bag (to distinguish them as non PVC subs). Catheters used on VCL6 are different to catheters used in other machines as they are non PVC catheters with non PVC funnels more commonly known as TPE. Please ensure to scan WIP ticket into machine and SAP to ensure correct catheters are used. Catheters must be placed back into the original blue bags after changeovers etc.

Each blue bag of catheters has a WIP Ticket attached which contains the relevant details i.e. part number, machine number, Lot number, the date, shift and the quantity made.

Note: Materials status is clearly marked by a green "QA acceptance" Stamp. Only materials with this stamp may be used.

Catheter Subs come in different widths and lengths.

VCL6 has the ability to coat leapfrog catheters such as Tiemann, Onli and Infyna catheters. Infyna funnels have identifiable grips on the catheters and the female product is a pastel colour.

| Name | Size |
|---|---|
| CH08 | 20cm, 17cm and 40cm |
| CH10 | 20cm, 17cm and 40cm |
| CH12 | 20cm, 17cm and 40cm |
| CH14 | 20cm, 17cm and 40cm |
| Ch16 | 40cm |

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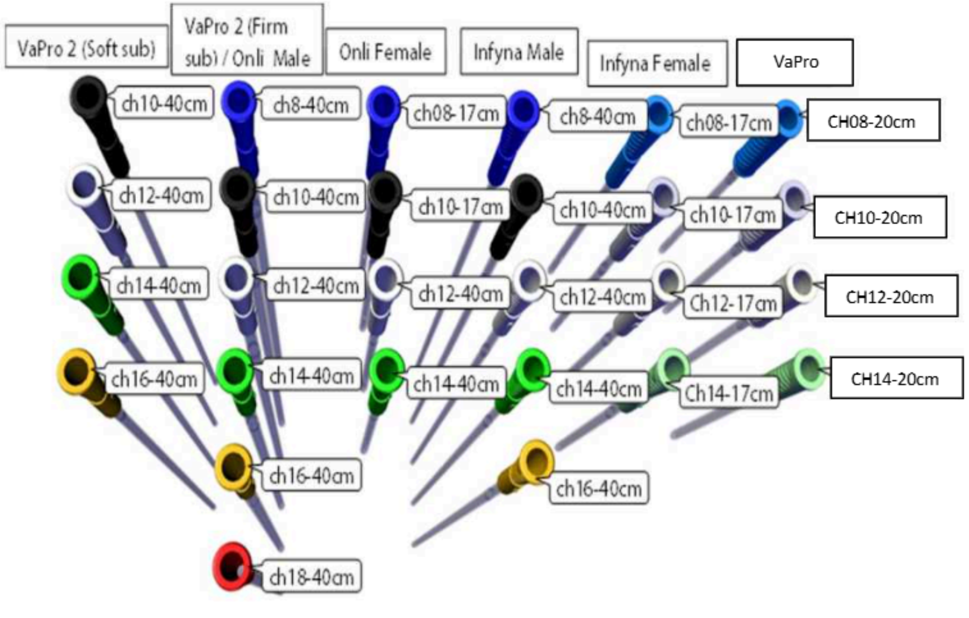
| MATERIALS FOR VAPRO COATING LINE 6 (VLC6)<br><br><br><br>Female Infyna funnels |  |
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## MATERIALS FOR VAPRO COATING LINE 6 (VLC6)

Ethanol 99% – Is stored in an IBC container and is located in Chemstore 2. This is used to clean each Catheter Sub.


The Ethanol is fed via the **'FMS'** (Fluid Management System) direct into the **VCL6** machine.

LCC (Steel Drums) – is stored in Chemstore 2 and is fed via the **'FMS'** (Fluid Management System) direct into the VCL Dip station.

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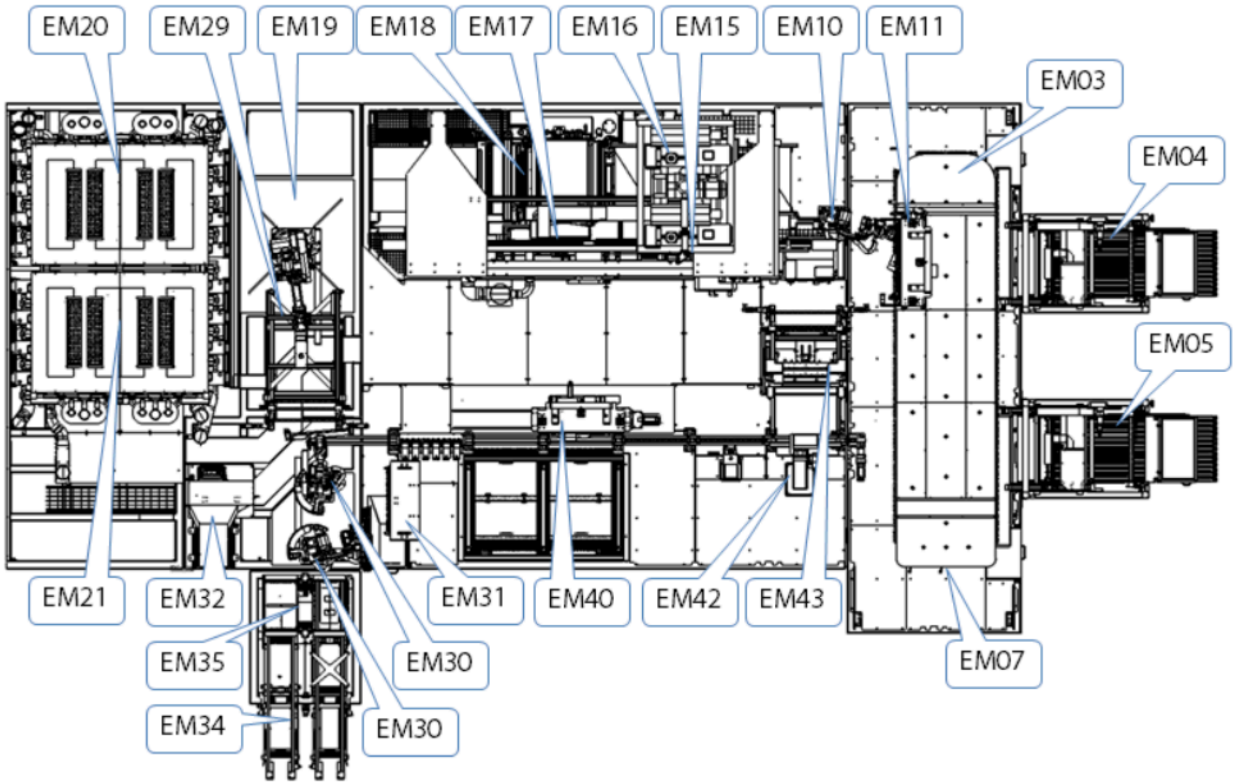
## MACHINE OVERVIEW




*EM- Equipment Module

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## MACHINE OVERVIEW

Main Transport (Station EM03) - This station consists of several small carriages/Jigs that ride on a guide rail. Each carriage holds 12 catheters. There are a number of stations which have direct interaction with station EMO3.

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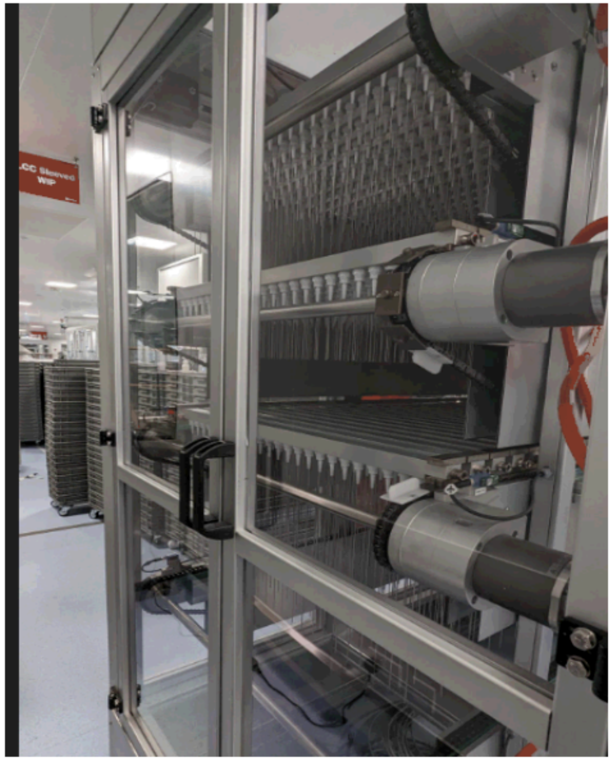
## MACHINE OVERVIEW - BUFFER CONVEYOR

The buffer conveyor stores 320 carriers with 24 mandrels on each carrier. It has four (4) levels each capable of storing 80 carriers.

A robot picks and places the carriers from the conveyor.

During a product changeover that requires the carriers to be changed according to the new recipe selected, the empty carriers are picked from VCL6 and placed into the Carrier Hotel (EM45). The new carrier picked from EM45 is then placed onto EM41 where EM30 places them into VCL6 via EM40.

The carriers are then checked for bent mandrels and leaks or blockages in the air purge valves before being placed on the carrier buffer ready to be loaded.



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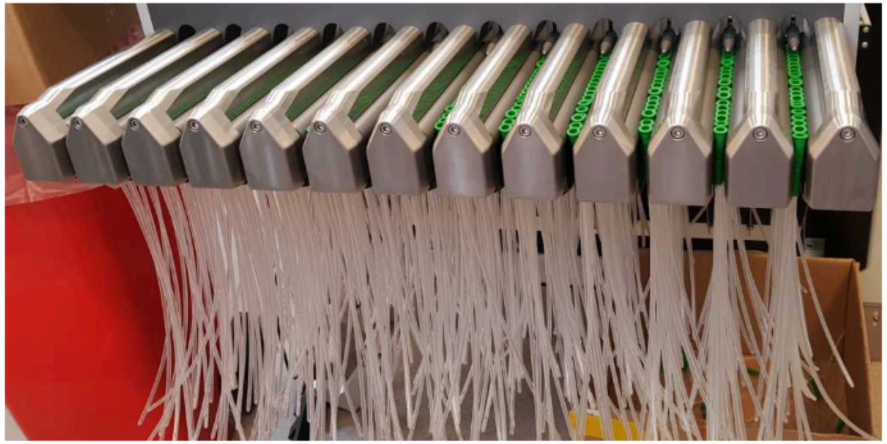
## MACHINE OVERVIEW – UNIT 1

**Catheter Load Station / Manual Infeed 1 & 2 (Stations EM04 and EM05)** - The Manual Load station has two sets of loading tracks that are hand-fed by operators, EM04 and EM05.
Each station has 12 loading tracks which can hold up to 2000 catheters.

There are 12 lanes on each manual Infeed station, catheters are fed down the lanes and into Dead nests to be picked by two Pick & Place Units. The Pick & Place Units place the catheters onto the Infeed Carousel EM03.

If 1 or more of the 12 catheters is not detected on the Pick & Place unit the machine automatically unloads the catheters onto the recycle belt which places the catheters in a recycle bin to be manually re fed back into the lanes.

Once the catheters are loaded onto the Infeed Carousel they must pass through a colour detection check EM07.



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## MACHINE OVERVIEW – UNIT 1

**Colour Inspection/Quality inspection (Station EM07)** - At the colour inspection station a sensor checks the colour is correct for each catheter e.g. green for ch14.

This check ensures the colour of the catheter is correct for 2 reasons:

1. Operator is using the correct SKU
2. To ensure that the catheter does not cause damage to the mandrels at the next stage e.g. a CH10 catheter would cause damage to a CH16 mandrel.

If the catheters do not pass the colour inspection check the machine will stop and the operator must manually remove all 24 catheters, check on the HMI which cathether(s) failed the colour detection aswel as visually inspecting the catheters, discard the bad parts and re use the good parts.

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## MACHINE OVERVIEW – UNIT 1

**Mandrel Insertion Station (Station EM11)** - The Fanuc robot takes empty carriers from the Carrier buffer station EM43 and presents them to the Mandrel insertion station. The Mandrel insertion station loads the 24 catheters onto the carrier, sensors here check the length of the catheters before inserting the mandrels to ensure the correct catheter length has been loaded, if the sensors detect a short/long part the machine will fault, the carrier will be placed on EM11 reject station and the operator must remove this carrier from the machine and place it on the carrier inspection trolley.

The fully loaded carrier is then transferred by a robot to the Plasma Park station of Unit 2

**A bump cap is mandatory to wear when entering VCL6 to retrieve carriers.**

Carrier reject station and light curtain.

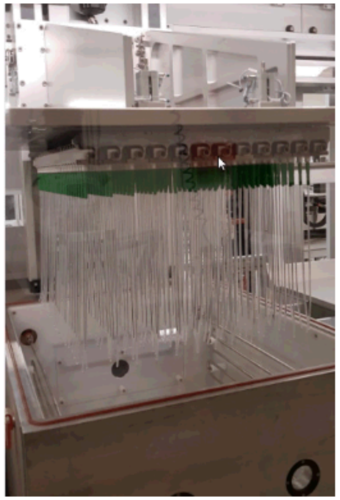
## MACHINE OVERVIEW – UNIT 2

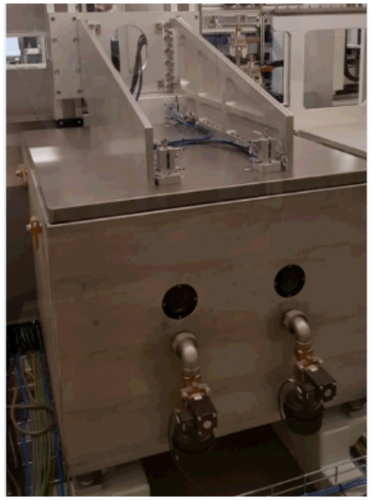
**Pick and Place Plasma (Station EM15) and Plasma Station (Station EM16)** - Once the 12 carriers (288 catheters) are placed on the plasma park station the pallet is picked up and placed into the plasma chamber, the head of the Pick & Place Unit acts as a lid that seals to the Plasma chamber.
A vacuum is pulled in the chamber, once the vacuum set point is reached the chamber stabilises with addition of atmospheric air and the plasma treatment is completed for a set time period, plasma is used to give a uniform surface activation to the catheters, this will allow the coating to adhere properly.

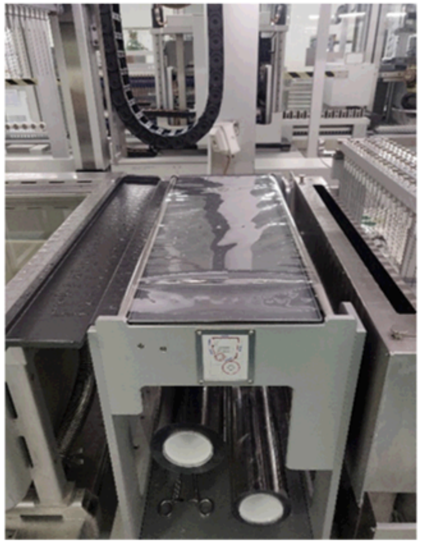
The chamber is then vented and the pallet is removed from the Plasma treatment station and placed onto the Dip Park station.

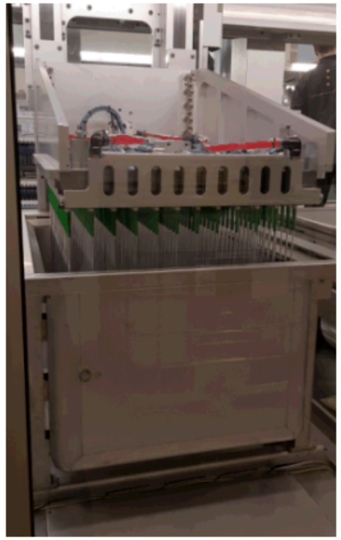
**Top Coat Station (Station EM17A)** - The pallet is then picked up from the dip park station and dipped into a tank with LCC top coat. The air in the catheters is trapped during dipping to prevent coating travelling up the lumen of the catheter via the eyelets. For punched catheters, coating within the lumen is removed via an air purge when the pallet has been lifted from the LCC coating.

**Dap station (Station EM18)**
Once the pallet has been lifted from the LCC coating it is moved on to the dap station where the tip of the catheters is lightly dapped on to foil known as dap foil to remove any excess coating the foil then rotates around an empty roll once it has been used.









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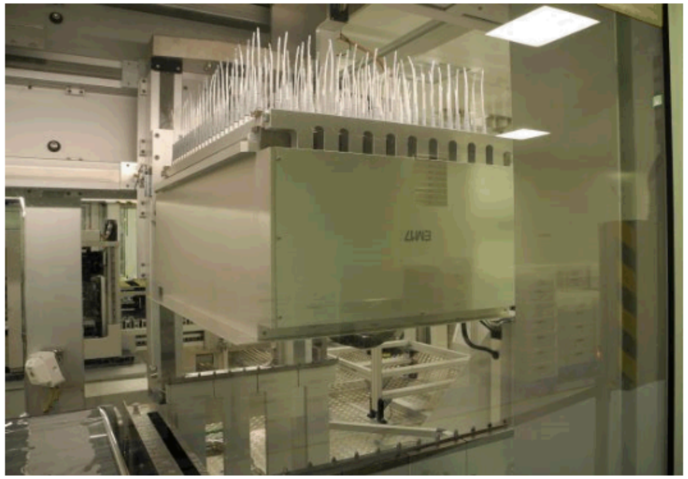
## MACHINE OVERVIEW – UNIT 2

**Flip station (Station EM18)** - The pallet is then flipped to 180 degrees for a set time to ensure coating is evenly distributed and there is no latent drip off the tip of the catheter



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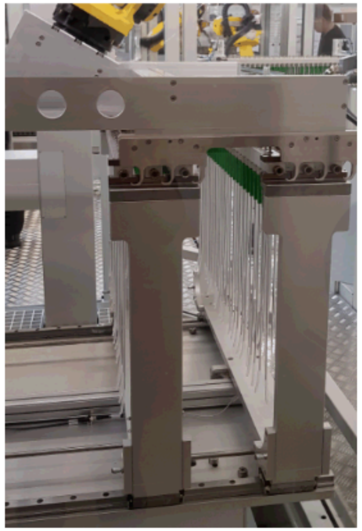
## MACHINE OVERVIEW – UNIT 2

**UV Robot Separation (EM19A) and Cure Chambers (EM20 & EM21)** - The pallet then gets placed on the separation buffer where it gets divided into 2 sets of 144 catheters. The 2 sets of 144 catheters are placed into a pair of UV chambers at the same time.

There are 2 pairs of UV chamber, 4 chambers in total, each with a dedicated airflow and temperature control system.

The airflow circulates continuously and UV exposure to cure the Top Coat begins immediately after the product has been placed into the chambers and the blinds close. The UV shutters close once the selected UV dose has been reached but the product remains in the chamber for a set time to dry the Top Coat. The catheters are then removed from the chambers and placed on the Outfeed Park station (EM29) by the fanuc robot.



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## MACHINE OVERVIEW – UNIT 2

**Buffer Conveyor (EM29) -** After the catheters have been cured in the UV chambers the carriers then get placed on the buffer conveyor, the robot carrier outlet (EM30) takes one carrier at a time and places it on a belt (EM31) where the catheters are removed from the mandrels, the catheters then will be either placed into the final reject bin if the pallet is being rejected for something that went wrong during the process or loaded into grey production trays.

**Note:** If something goes wrong with the process- e.g. plasma park time max out then all 288 catheters on the pallet get rejected and placed into the reject bin.

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| | |
|---|---|
| **MACHINE OVERVIEW – UNIT 2**<br><br>Carrier Management - When new or repaired carriers are introduced into the machine they must be placed on top of EM36, the EM44 robot will then pick up the carriers and place them on EM41 Conveyor. | |

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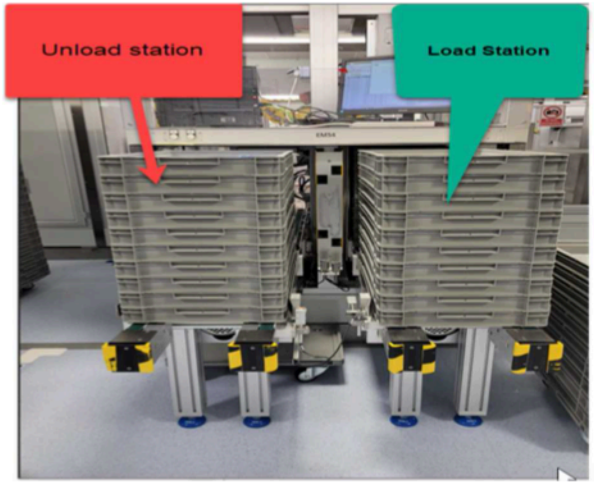
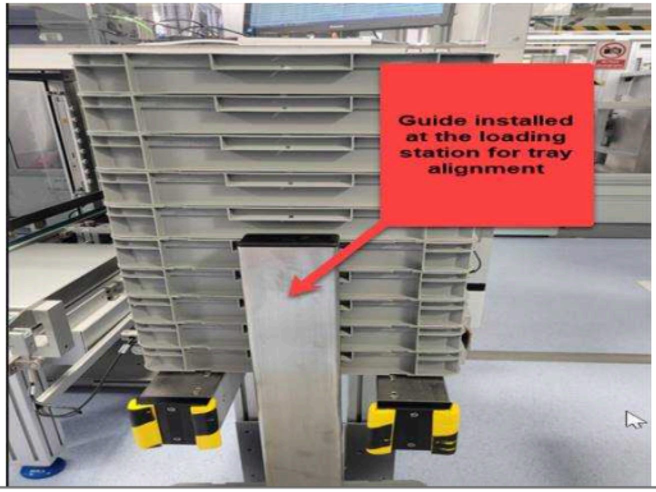
## MACHINE OVERVIEW – UNIT 2

**Offload - EM34:**
EM 34 has a tray-unloading conveyor to the left-hand side and a loading station to the right-hand side of the station.
Please use the guide installed at the machine at EM 34 on the right-hand to align trays prior to acknowledging drop-off. This is important for the correct alignment of the trays to prevent a crash at the swivel plate.

**Load & unload stations:**




**Machine Guide :**



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## HMI OVERVIEW

VCL6 have three HMI screens, one at the infeed station, one at the outfeed station and one located at the door to the main module- as seen in the drawing.

Each HMI can control the whole machine once you press the HMI control button.

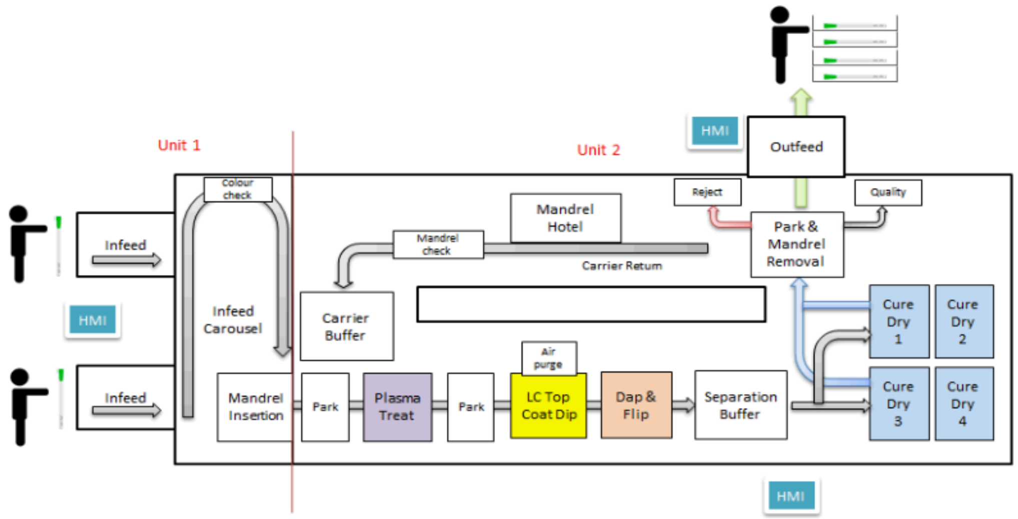
**Each HMI screen has the following buttons**

**HMI CONTROL**

Each HMI has a request control button- this is a safety feature to ensure that only one HMI can be used at a time
Before access is granted to a HMI, the user must press the HMI control button, when pressed the button will blink for 3 seconds and then go solid white if none of the same buttons on the 2 other HMI's have been pressed.

When button is solid white you have control over HMI. All other HMI's are blocked.

If the button with solid white light starts to blink, someone else wants control over HMI. To keep control you have to press the button to cancel their request.



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## HMI OVERVIEW

### Start Button

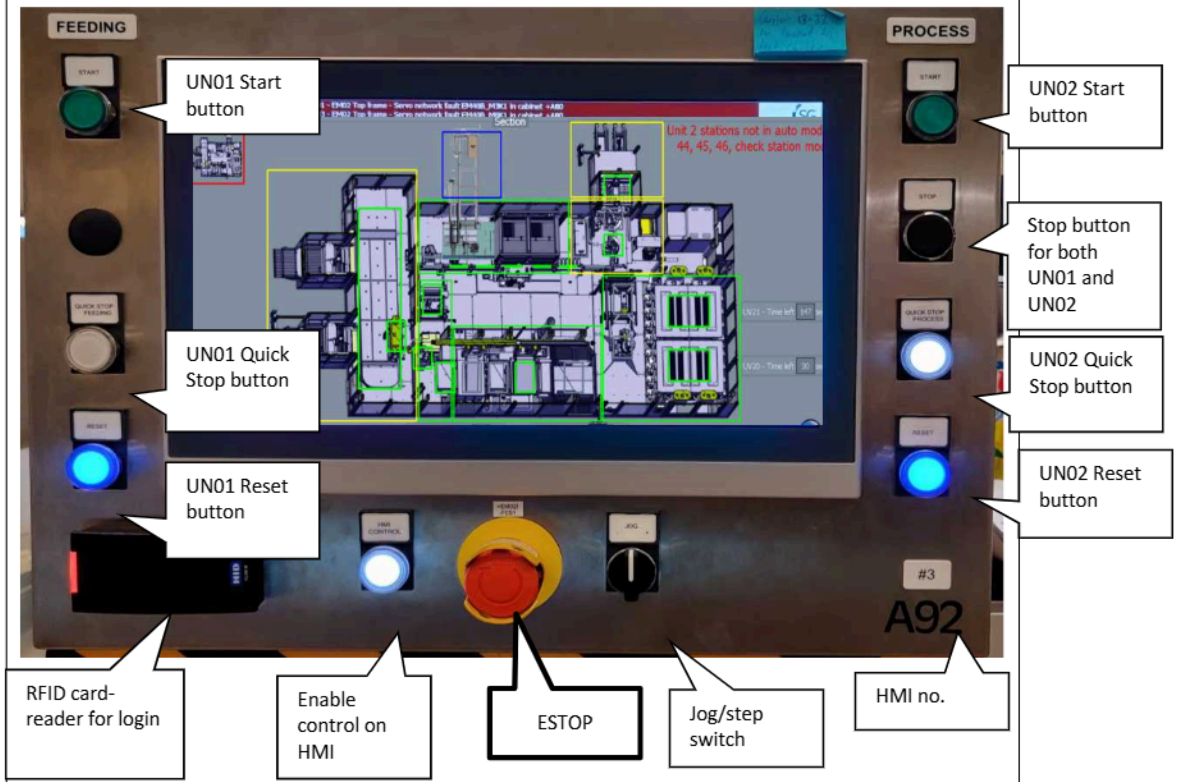
The start button is green. It flashes when the machine is in state Resetting and illuminates continuously when in state Running, there is a start button for both feeding and process to control which unit of the machine you wish to start

### Stop button

The controlled stop button is black and located on the right hand side of the HMI's. When pressed, the state changes to first Stopping and when it has taken care of all products so no pallets gets rejected, it will go to state Stopped, this button will stop both units of the machine.

### Quick Stop button

The quick stop button is white. When pressed, the state changes to first Stopping and when all equipment is done with its processes it then enters state Stopped, there is a quick stop for both feeding and process to control which unit of the machine you wish to stop



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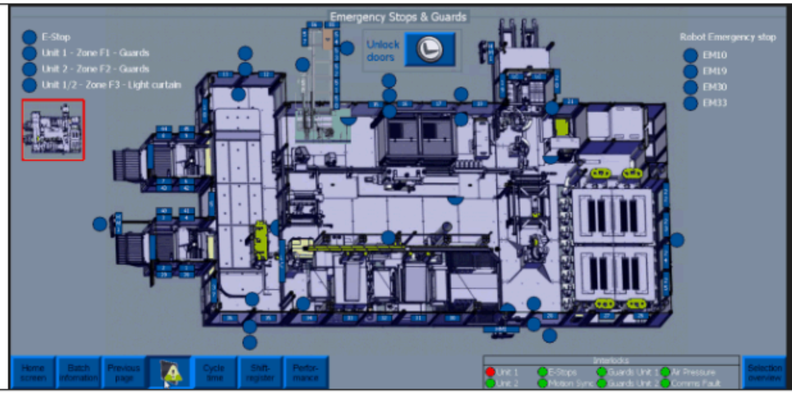
## HMI OVERVIEW

To unlock door of the carrier hotel, go to the safety overview screen, and the button can be seen at the top of the screen. Select this button to open doors on carrier hotel.



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## HMI OVERVIEW

**Jog/step button**

The Jog button is a black rotary switch and has three positions. It automatically return to its initial position, which is the center position. It can be turned clockwise towards "+" or counterclockwise towards "-". The purpose of the button is to jog control modules/components (e.g. a motor) in positive and negative directions and has the same functionallity as the Jog buttons on the screen. The jog function is only for manual operation.

**RFID card reader**

The RFID card reader can be used to log into the machine using a clockcard instead of logging on manually

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## END OF SHIFT

- At the end of a shift and/or end of a batch, the operator must press line empty on the HMI, wait for all catheters in process to be removed from the machine via the outfeed and fill in the relevant details to complete the production sheets 75CLR00611_Att2 and place them on the Group Leader's desk.
- At the end of the shift, the operator must ensure to report the product count and the amount of calculated waste from the screen into the Hand Held Terminal (HHT) (See how to report goods in the SAP – Training Guide for Production Operators 75GEN00060_Att6) and place the production paperwork it on the Group Leader's desk.
- The machine must be kept neat and tidy at all times.

**Note:** All surfaces of the VaPro Coating Line 6 Machine and associated work benches must be cleaned with 70% IPA at start-up and once more per shift as per 75CLR00611_Att2. A record of this cleaning must be recorded in the Production Sheet 75CLR00611_Att2.

To ensure that cleaning is carried out in the correct manner please follow the guidelines below;

- Always wear gloves to protect your skin before using IPA
- Always spray IPA onto the wipe, not the surface being cleaned - this will prevent IPA being sprayed into the eyes

Always ensure that you use enough IPA to wet the surface being cleaned.

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## CHANGEOVERS

When a changeover has occurred, the operator must ensure that the area is cleared of all components and the assemblies not required and are removed to the racks or the holding areas in the original blue bags.

When changing from one SKU to another the operator need to only remove the catheters from the manual feed track as per changeover standard work ensuring to place product back into original blue bags. The operator must complete a physical check to ensure no catheters remain on the machine.

To prevent cross contamination, a line clearance must be performed prior to a changeover.

The Line Clearance for VCL6 Machine consists of:

1. Checking each compartment/station for rogue catheters, etc.
2. Check the load stations for rogue catheters, assemblies etc.

**A record of the Line Clearance must be recorded in the production sheet 75CLR00611_Att2. for VaPro Coating Line**

**Note:** When a work order has been finished, the operator must ensure that Start Up line Clearance, Verification Check and End of Procedure sections are filled out correctly on the work order sheet before placing it on the Team Leaders (TL) desk.

**Note**: All materials that are removed from the machine must have the original WIP (Work In Progress) ticket reapplied and must be in a blue bag. If the original WIP ticket is not available inform the CTL or GL immediately. If a material is removed at any other time it must have the original WIP ticket reapplied and be placed into the original blue bag.